

Latency and Compressibility: Two Independent Routes to Predictive Architecture in Adaptive Systems

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Abstract

This paper identifies two logically independent pressures that each, on its own, forces adaptive systems toward predictive architectures—that is, toward maintaining internal generative models of the environment rather than relying on reactive stimulus–response mappings.

The *temporal pressure* arises from finite signal propagation speed: any system with nonzero response latency operating in a changing environment must act on states that no longer obtain by the time action takes effect. Compensation requires that internal states carry information about the future beyond what is available in the present.

The *informational pressure* arises from the algorithmic complexity of the environment: when the environment cannot be compressed into a short description—when its Kolmogorov complexity is high relative to channel capacity—faithful reactive transmission of raw sensory states becomes impossible under Shannon’s channel constraints. The only recourse is to construct an internal model that encodes predictive regularities, transmitting compact model parameters rather than full state descriptions.

We formalize both pressures using information theory and algorithmic information theory, prove that each independently entails the necessity of predictive

internal states, and show that their convergence on the same architectural solution constitutes evidence for a deep structural constraint rather than a contingent design preference. We discuss implications for neuroscience, artificial intelligence, and the philosophy of adaptive systems, and connect the formal results to recent computational and neurobiological work on delay compensation, spiking neural codes, and the free energy principle.

Keywords: predictive processing, Kolmogorov complexity, Shannon information, neural latency, forward models, channel capacity, algorithmic information theory, free energy principle, active inference

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1 Introduction

A growing consensus across neuroscience, cognitive science, and artificial intelligence holds that brains and other adaptive systems operate as prediction machines—generating anticipatory models of the environment and updating those models in light of incoming data [1, 5, 21, 25]. This idea, broadly called *predictive processing*, has been independently rediscovered in motor control [4, 26], perception [8, 11], evolutionary neuroscience [19], and machine learning [14].

Despite its empirical success, predictive processing is often treated as one computational strategy among several—a framework that happens to fit much of the data, but that might in principle be replaced by alternatives such as direct perception [6], purely feedforward processing [16], or reactive control. The question of whether prediction is *necessary*—whether the physics and mathematics of adaptive systems leave any viable alternative—has received less formal attention.

A companion paper [13] argued that physical latency constraints make predictive architecture the only evolutionarily stable solution for systems operating in dynamic, punishing environments. That argument proceeds from a single source of pressure: the finite speed of neural signal propagation creates a temporal gap between the world’s current state and the system’s ability to respond, forcing anticipation.

The present paper extends this analysis by identifying a *second*, logically independent pressure toward the same architectural conclusion. This second pressure arises not from temporal delay but from the *informational structure* of the environment. When the environment has high algorithmic complexity—when it cannot be compressed into a short description—a reactive system that attempts to transmit full state information through a finite-capacity channel will fail, regardless of how fast that channel operates. The only solution that remains viable under bandwidth constraints is to compress the environment into a predictive model: a compact internal representation that captures regularities and discards noise.

The central thesis is therefore:

Temporal latency and environmental complexity are two independent forces, each of which, acting alone, is sufficient to rule out purely reactive (memory-less) architectures. Their convergence on the same requirement—that internal states must carry information about the future—constitutes evidence that some form of predictive capacity is a structural constraint on adaptive systems operating under realistic physical conditions, not a contingent design choice.

A note on scope is warranted at the outset. The formal results below establish that *purely reactive* systems—those whose internal states carry zero information about the future beyond what the present observation provides—face exponential survival decay.

This rules out the memoryless end of the architectural spectrum. It does not, by itself, mandate a specific implementation such as a hierarchical generative model. Between the memoryless reactive extreme and a full generative world model lies a continuum of architectures—systems with memory buffers, cached policies, error-correcting codes, reservoir dynamics—many of which satisfy our formal criterion $\mathcal{I} > 0$ to varying degrees. The framework identifies the minimal requirement (nonzero predictive information) but does not prescribe how richly that requirement must be met. We return to this issue in Section 10.

The paper is organized as follows. Section 2 defines the formal framework. Section 3 develops the temporal pressure argument. Section 4 develops the informational pressure argument from Kolmogorov complexity and Shannon theory. Section 5 proves the convergence theorem. Section 6 presents a toy model with specified dynamics that demonstrates the phase boundary and validates the formal results by simulation. Section 7 characterizes the phase boundary between reactive and predictive regimes. Section 8 connects the formal results to neuroscience. Section 9 discusses implications for artificial intelligence. Section 10 addresses objections. Section 11 situates the results within existing frameworks. Section 12 concludes.

2 Formal Framework: Systems, Environments, and Channels

Before developing either argument, we establish a common formal language. The definitions below are stated precisely but are accompanied by commentary to ensure the reasoning remains transparent to readers across disciplines.

2.1 Adaptive Systems

Definition 1 (Adaptive System). An *adaptive system* is a tuple $\mathcal{S} = (X, A, U, \tau, C)$ where:

- X is the set of internal states of the system,
- A is the set of actions available to the system,
- $U : X \rightarrow A$ is the update rule mapping internal states to actions,
- $\tau > 0$ is the response latency—the minimum time between receiving a sensory input and producing an action,
- $C > 0$ is the channel capacity—the maximum rate (in bits per second) at which the system can transmit information from sensors to its action-selection mechanism.

This definition captures any entity—a bacterium, a brain, a robot, a trading algorithm—that senses its environment, maintains some internal state, and produces actions after

some delay and through a channel of limited bandwidth. The latency τ reflects the physical fact that signals propagate at finite speed and that computation takes time. The channel capacity C reflects the physical fact that any real communication channel—including neural pathways—can transmit only a bounded amount of information per unit time, a limit established by Shannon’s noisy-channel coding theorem [22].

2.2 Environments

Definition 2 (Environment). An *environment* is a tuple $\mathcal{E} = (Y, P, V, \kappa)$ where:

- Y is the set of environmental states,
- P is a transition kernel: $P(Y_{t+\Delta} \mid Y_t)$ gives the probability distribution over future states given the present state,
- $V \subset Y$ is the *viability region*—the set of environmental states in which the system can persist,
- $\kappa \in (0, 1]$ is the *punishment severity*—the probability that a single mismatched action causes the system to leave V .

The viability region V captures the idea that organisms must remain within certain bounds to survive: a fish must stay in water, a cell must maintain osmotic balance, a self-driving car must stay on the road. The punishment parameter κ measures how unforgiving the environment is: in a predator–prey encounter κ may be close to 1 (a single error is fatal), while in a benign laboratory setting κ may be small.

2.3 Environmental Volatility

The rate at which the environment changes is central to both arguments. We measure this with conditional entropy.

Definition 3 (Environmental Volatility). The *volatility* of environment $\mathcal{E}$ at timescale τ is:

$$H_\tau := H(Y_{t+\tau} \mid Y_t), \tag{1}$$

the conditional Shannon entropy of the future state given the present state, measured in bits.

Entropy, introduced by Shannon [22], quantifies uncertainty: $H_\tau = 0$ means the future is fully determined by the present (a frozen or perfectly periodic environment), while large H_τ means the future is highly uncertain given current information. Most real environments fall between these extremes—partially predictable, partially stochastic. The subscript τ makes explicit that volatility depends on the timescale: an environment that is predictable over milliseconds may be unpredictable over hours.

2.4 Algorithmic Complexity of the Environment

Beyond statistical unpredictability, environments also vary in their *structural complexity*—how much description is required to capture their regularities.

Definition 4 (Kolmogorov Complexity). For a string s describing an environmental state sequence, the *Kolmogorov complexity* $K(s)$ is the length (in bits) of the shortest program on a universal Turing machine that outputs s and then halts [12, 15].

Kolmogorov complexity captures the intrinsic compressibility of a data stream. A sequence of all zeros has very low Kolmogorov complexity (a short program can generate it), while a truly random sequence has Kolmogorov complexity approximately equal to its own length—it cannot be compressed at all.

For an adaptive system, this matters because if the environment’s state sequence has high Kolmogorov complexity, then there is no short description that captures the raw data. Any system attempting to transmit or store full environmental state descriptions must handle data volumes that grow proportionally to the complexity of the environment. When this volume exceeds the system’s channel capacity, faithful reactive processing becomes impossible.

Definition 5 (Effective Complexity Rate). The *effective complexity rate* of the environment is:

$$\rho := \lim_{T \rightarrow \infty} \frac{K(Y_1, Y_2, \dots, Y_T)}{T}, \quad (2)$$

the asymptotic Kolmogorov complexity per time step of the environmental state sequence, measured in bits per step.

When ρ is large relative to channel capacity C (adjusted for the system’s sampling rate), the environment is too complex to relay faithfully—the system cannot transmit raw sensory states to its decision-making core quickly enough for reactive control. This is the informational bottleneck that drives the second argument.

2.5 Predictive Capacity

We now define what it means for a system to be predictive.

Definition 6 (Predictive Capacity). The *predictive capacity* of system $\mathcal{S}$ with respect to environment $\mathcal{E}$ at timescale τ is:

$$\mathcal{I} := I(X; Y_{t+\tau} \mid Y_t), \quad (3)$$

the conditional mutual information between the system’s internal states and the future environmental state, given the present environmental state.

Mutual information, a cornerstone of information theory, measures the amount of information that one random variable carries about another. The conditional form $I(X; Y_{t+\tau} \mid Y_t)$ specifically measures the *additional* information that the system’s internal states X carry about the *future* environmental state $Y_{t+\tau}$, beyond what is already available from observing the *present* state Y_t .

If $\mathcal{I} = 0$, the system’s internal states tell us nothing about the future that the present observation does not already reveal. Such a system is purely reactive: it can only respond to the current state. If $\mathcal{I} > 0$, the system has encoded some information about where the environment is heading—it is, by definition, predictive.

This definition is deliberately broad. It encompasses a frog’s retinal ganglion cells extrapolating a fly’s trajectory, a cerebellum generating predicted limb positions, an orca’s neural circuits anticipating prey evasion paths, and a language model’s hidden states encoding likely continuations of a text. What unites these cases is the formal structure: internal states carrying information about states not yet directly observed.

3 The Temporal Pressure: Latency Forces Prediction

The first route to predictive architecture proceeds from the physics of signal propagation. This argument was developed in detail in Kriger [13]; here we present a streamlined formal version.

3.1 The Latency Gap

Every adaptive system with $\tau > 0$ faces a fundamental mismatch: the action it selects at time t takes effect only at time $t + \tau$. If the environment has changed during that interval—and if the environment is volatile, it will have—then the action is calibrated for a state that no longer obtains.

Proposition 7 (Latency-Induced Error Bound). *For a purely reactive system ($\mathcal{I} = 0$) in a non-stationary environment ($H_\tau > 0$) with punishment severity $\kappa > 0$ and state space cardinality $|Y|$, in the regime of substantial volatility ($H_\tau \gg 1$), the per-step probability of a viability-threatening error is bounded below:*

$$p_{\text{err}} \geq \kappa \cdot \frac{H_\tau}{\log |Y|}. \quad (4)$$

More precisely, Fano’s inequality yields $p_{\text{err}} \geq \kappa \cdot \frac{H_\tau - 1}{\log |Y|}$, which reduces to the bound above when $H_\tau \gg 1$. The qualitative conclusion—that $p_{\text{err}} > 0$ whenever $H_\tau > 0$ and $\kappa > 0$ —holds without the asymptotic restriction.

The quantity $\frac{H_\tau}{\log |Y|}$ represents the fraction of the state space that is uncertain at timescale τ , normalized to the range $[0, 1]$.

Proof. A reactive system selects actions based solely on the present observed state Y_t , since by assumption $I(X; Y_{t+\tau} | Y_t) = 0$. The action is therefore optimal for state Y_t . The probability that the action is mismatched to the actual state $Y_{t+\tau}$ when it takes effect is at least $P(Y_{t+\tau} \neq Y_t)$. By Fano's inequality [2], for a random variable $Y_{t+\tau}$ taking values in a set of size $|Y|$, the probability of error in estimating $Y_{t+\tau}$ from Y_t alone satisfies:

$$P(Y_{t+\tau} \neq Y_t) \geq \frac{H(Y_{t+\tau} | Y_t) - 1}{\log |Y|}.$$

This bound is tight when $Y_{t+\tau}$ is approximately uniformly distributed over the uncertain states given Y_t ; it becomes loose when the conditional distribution is highly skewed. Since each mismatch causes viability failure with probability κ , the per-step viability-threatening error rate satisfies the exact Fano-based bound $p_{\text{err}} \geq \kappa \cdot (H_\tau - 1) / \log |Y|$. In the regime $H_\tau \gg 1$, the unit correction is negligible, yielding (4). For the qualitative conclusion ($p_{\text{err}} > 0$), we note that $H_\tau > 0$ implies $P(Y_{t+\tau} \neq Y_t) > 0$ directly, without requiring Fano's inequality. $\square$

3.2 Exponential Survival Decay

A nonzero per-step error probability compounds over time.

Theorem 8 (Temporal Survival Decay). *For a purely reactive system ($\mathcal{I} = 0$) in a non-stationary, punishing environment, the probability of surviving n steps decays exponentially:*

$$P(\text{survive } n \text{ steps} | \mathcal{I} = 0) \leq (1 - p_{\text{err}})^n \rightarrow 0 \quad \text{as } n \rightarrow \infty. \quad (5)$$

This result captures a fundamental fact: if a system makes errors at a constant nonzero rate and each error carries a chance of termination, eventual extinction is certain. The only question is how quickly. For a reactive system in a volatile, punishing environment, survival probability decreases geometrically with each time step.

Proof. Under the assumption that error events across steps are at least conditionally independent (which holds when the environment's transition kernel is Markov and the system has no memory, i.e., $\mathcal{I} = 0$), the probability of surviving n consecutive steps is $\prod_{i=1}^n (1 - p_{\text{err},i}) \leq (1 - p_{\text{err}})^n$, where $p_{\text{err}} = \min_i p_{\text{err},i} > 0$ by Proposition 7. Since $0 < 1 - p_{\text{err}} < 1$, this product converges to zero. $\square$

3.3 Prediction as Temporal Compensation

A system with $\mathcal{I} > 0$ can reduce the effective error rate. If the system's internal states carry information about $Y_{t+\tau}$ beyond what Y_t reveals, actions can be calibrated not for the present state but for the anticipated future state.

Proposition 9 (Predictive Error Reduction). *A system with predictive capacity $\mathcal{I} > 0$ achieves a per-step error probability:*

$$p_{\text{err}}(\mathcal{I}) \leq \kappa \cdot \frac{H_\tau - \min(\mathcal{I}, H_\tau)}{\log |Y|}. \quad (6)$$

Each bit of predictive information $\mathcal{I}$ reduces the effective uncertainty the system faces. When $\mathcal{I} \geq H_\tau$, the system has enough predictive capacity to fully compensate for environmental volatility, and the error term vanishes. This formalizes the intuition that prediction bridges the latency gap: by encoding information about the future, the system acts as though its effective latency were zero.

The result has been demonstrated computationally in the domain of oculomotor control by Perrinet et al. [18], who showed that uncompensated sensorimotor delays in smooth pursuit eye movements produce catastrophic oscillatory instability. Their solution employs representations in *generalised coordinates of motion*—encoding not just the current position of a target but also its velocity, acceleration, and higher derivatives—which enables the system to project delayed sensory signals forward in time. The generalised coordinates carry conditional mutual information $I(X; Y_{t+\tau} \mid Y_t) > 0$, enabling delay compensation.

Similarly, Khoei et al. [11] demonstrated that the flash-lag effect in visual perception can be explained by a Bayesian model that uses velocity estimates to extrapolate delayed sensory information to the present—a mechanism they term *parodiction*. The precision of the velocity estimate modulates the magnitude of the predictive shift, corresponding formally to the magnitude of $\mathcal{I}$ in our framework.

4 The Informational Pressure: Complexity Forces Compression into Models

We now develop the second, independent argument. Even if latency were zero—even if a system could respond instantaneously—there is a separate reason why reactive architectures fail when environments are complex.

4.1 The Bandwidth Bottleneck

Every physical system processes information through channels of finite capacity. Neurons transmit spikes at bounded rates. Axons carry signals at finite speeds with finite bandwidth. Sensory organs sample the environment at limited spatial and temporal resolutions. These are not engineering limitations to be overcome; they are physical constraints inherent to any material information-processing system.

Shannon’s channel capacity theorem [22] establishes that a channel with capacity C (measured in bits per second) cannot reliably transmit information at a rate exceeding C . Any attempt to do so results in unavoidable information loss.

For a reactive system—one that attempts to relay raw sensory states to its action-selection mechanism without compression—the required transmission rate is determined by the complexity of the environment.

Definition 10 (Reactive Transmission Demand). The *reactive transmission demand* of environment $\mathcal{E}$ sampled at rate f_s (samples per second) is:

$$\mathcal{D} := f_s \cdot \rho, \quad (7)$$

where ρ is the effective complexity rate (Definition 5), measured in bits per second.

This quantity represents the data rate required for lossless reactive transmission: the system must transmit ρ bits of incompressible information per time step, and it samples f_s times per second, yielding a total demand of $\mathcal{D}$ bits per second.

4.2 The Channel Overflow Condition

When the reactive demand exceeds the channel capacity, faithful reactive processing is impossible.

Proposition 11 (Channel Overflow). *When $\mathcal{D} > C$, a reactive system necessarily loses information at rate:*

$$\mathcal{L} = \mathcal{D} - C > 0 \quad \text{bits per second.} \quad (8)$$

This information loss $\mathcal{L}$ is not a matter of engineering quality—it is a mathematical consequence of Shannon’s theorem. No encoding scheme, no matter how clever, can transmit $\mathcal{D}$ bits per second through a channel of capacity $C < \mathcal{D}$ without losing at least $\mathcal{D} - C$ bits per second.

Proof. By the converse of Shannon’s noisy-channel coding theorem [2, 22], reliable transmission at rate $R > C$ is impossible: for any code operating above capacity, the probability of decoding error is bounded away from zero. Therefore, a reactive system attempting to transmit raw environmental state descriptions at rate $\mathcal{D} > C$ must incur information loss of at least $\mathcal{D} - C$ bits per second. $\square$

The consequence for an adaptive system is immediate: lost information means that the system’s action-selection mechanism receives an incomplete, degraded representation of the environment. Actions selected on the basis of this degraded representation will be mismatched to the true environmental state, producing errors with the same compounding dynamics described in Theorem 8.

4.3 The Compression Imperative

If raw reactive transmission fails, the only alternative is to compress. But compression of incompressible data—data with high Kolmogorov complexity—is, by definition, impossible without loss. This seems to create an impasse. The resolution lies in a critical distinction between two kinds of information in the environmental signal.

Definition 12 (Predictive–Stochastic Decomposition). The environmental state sequence can be decomposed into:

$$\rho = \rho_{\text{pred}} + \rho_{\text{stoch}}, \quad (9)$$

where ρ_{pred} is the *predictive complexity rate*—the rate of information that reflects exploitable regularities (temporal correlations, causal structure, repeating patterns)—and ρ_{stoch} is the *stochastic complexity rate*—the rate of genuinely random, structureless variation that cannot be predicted or compressed.

This decomposition mirrors a fundamental distinction in information theory. The total information in a data stream consists of *structure* (patterns, correlations, laws) and *noise* (random variation). A predictive model captures the structure and discards the noise. In doing so, it achieves a form of compression that is not merely quantitative (fewer bits) but qualitative (retaining the bits that matter for action while discarding those that do not).

The connection to Kolmogorov complexity is as follows. The predictive component ρ_{pred} corresponds to the regularities that a short program could capture—the compressible part of the environment. The stochastic component ρ_{stoch} corresponds to the incompressible residual. A reactive system must transmit both. A predictive system needs to transmit only updates to its model—which scale with ρ_{pred} , not with ρ .

4.4 The Model as Compressor

Theorem 13 (Informational Necessity of Prediction). *For an adaptive system with channel capacity C in an environment with reactive transmission demand $\mathcal{D} > C$ and predictive complexity rate ρ_{pred} such that $f_s \cdot \rho_{\text{pred}} \leq C$, the system can maintain viability only if it maintains an internal model M such that:*

$$I(M; Y_{t+\tau}) \geq H_\tau - \frac{C}{f_s}. \quad (10)$$

This theorem states that when the full environmental signal exceeds channel capacity but the predictive regularities fit within it, the system *must* build an internal model that captures those regularities. The model M carries sufficient mutual information with the future state $Y_{t+\tau}$ to compensate for the information that the channel cannot carry.

The internal model functions as a compressor: instead of transmitting a complete description of the current environmental state (which requires ρ bits per step), the system transmits only the *prediction errors*—the deviations between what the model predicted and what actually occurred. These prediction errors carry at most ρ_{stoch} bits per step if the model is well calibrated, which by assumption fits within the channel.

Proof. Let R_{eff} denote the effective information rate the system can transmit about environmental states. By Shannon’s theorem, $R_{\text{eff}} \leq C$. For viability, the system must select actions that are calibrated to the true state $Y_{t+\tau}$ with sufficient accuracy. By Fano’s inequality, the probability of action–state mismatch is bounded by a decreasing function of the mutual information $I(\hat{Y}_{t+\tau}; Y_{t+\tau})$, where $\hat{Y}_{t+\tau}$ is the system’s estimate.

Without an internal model, $\hat{Y}_{t+\tau}$ depends only on what passes through the channel, so $I(\hat{Y}_{t+\tau}; Y_{t+\tau}) \leq C/f_s$ per step. If this is insufficient for viability (because $H_\tau > C/f_s$), the system must supplement channel-transmitted information with internally generated information—an internal model M such that $I(M; Y_{t+\tau})$ fills the gap. The minimum model quality required follows from requiring total available information to meet the viability threshold:

$$\frac{C}{f_s} + I(M; Y_{t+\tau}) \geq H_\tau,$$

which yields (10). □

Two features of this result deserve explicit comment. First, the theorem requires that the predictive regularities of the environment fit within the channel: $f_s \cdot \rho_{\text{pred}} \leq C$. This assumption does significant work. If the environment’s predictive structure itself exceeds channel capacity—if even the regularities are too rich to transmit—then no single-level model suffices. In this regime, the framework suggests (but does not formally prove) that a *hierarchy* of increasingly compressed models would be required, each level abstracting the level below. This connects naturally to hierarchical predictive coding architectures [21], but a full treatment of the hierarchical case is beyond the present scope.

Second, the decomposition $\rho = \rho_{\text{pred}} + \rho_{\text{stoch}}$ (Definition 12) is presented as a clean partition, but in practice it is *model-relative*: what counts as “predictive regularity” depends on the class of models the system can entertain. A system with a richer model class will identify more structure in the environment as predictive; a system with a poorer model class will classify more of the same signal as stochastic noise. This means that ρ_{pred} and ρ_{stoch} are not intrinsic properties of the environment alone but joint properties of the environment–model pair. The theorem should therefore be read as conditional: *given* a model class rich enough to capture the environment’s predictable structure within the channel, viability requires deploying such a model. The question of which model class suffices for a given environment is a separate problem—one that evolution and learning address through trial and error.

4.5 An Illustrative Example

To ground this result in a concrete scenario, consider an orca hunting a school of fish in turbid water using echolocation. The environment—the three-dimensional positions, velocities, and evasive trajectories of dozens of fish—has high Kolmogorov complexity at millisecond resolution. The orca’s acoustic channel (emitting clicks, receiving echoes, processing in auditory cortex) has a finite bandwidth determined by click repetition rate, cochlear frequency resolution, and neural processing capacity.

The reactive demand $\mathcal{D}$ —the bits per second needed to faithfully represent the full state of the prey school—far exceeds the channel capacity C of the echolocation system. A purely reactive orca, attempting to act based solely on each echo without an internal model, would face overwhelming information loss: most of what it needs to know about the fish would simply not make it through the channel.

The solution is a predictive model: the orca’s brain maintains an internal representation of the prey school’s dynamics—likely positions, probable evasion trajectories, anticipated convergence points. Each new echo does not need to carry the full state description; it only needs to carry the *difference* between what the model predicted and what actually occurred. This prediction-error signal is much lower bandwidth than the full state signal, fitting within the channel.

This is not a metaphor. It is a consequence of the informational constraint. The orca’s brain *must* be predictive not because prediction is a desirable cognitive feature, but because the physics of information transmission leaves no alternative.

5 Convergence: Two Routes, One Destination

We now show that the temporal and informational pressures, though logically independent, lead to the same architectural requirement.

5.1 Independence of the Two Pressures

The temporal argument (Section 3) depends on $\tau > 0$ and $H_\tau > 0$. It is valid even if $C = \infty$: even a system with unlimited bandwidth still faces the latency gap.

The informational argument (Section 4) depends on $\mathcal{D} > C$. It is valid even if $\tau = 0$: even a system with instantaneous response still faces the bandwidth bottleneck when the environment is too complex to relay.

Proposition 14 (Independence). *The conditions $\{\tau > 0, H_\tau > 0, \kappa > 0\}$ and $\{\mathcal{D} > C, \rho_{\text{pred}} > 0\}$ are logically independent: neither set of conditions entails or is entailed by the other.*

Proof. We exhibit environments satisfying each set of conditions without the other.

Temporal pressure without informational pressure: Consider an environment with $H_\tau > 0$ (volatile) but low Kolmogorov complexity (e.g., a simple periodic environment with noise). A reactive system with high channel capacity $C \gg \mathcal{D}$ can transmit complete state descriptions without loss, but still faces the latency gap because $\tau > 0$ and the environment changes within the response window.

Informational pressure without temporal pressure: Consider a system with negligible latency ($\tau \approx 0$, as in a tiny insect’s nervous system) in a complex environment. Even though the system can respond almost instantly, if the environmental state descriptions exceed channel capacity, faithful reactive processing is still impossible—the bottleneck is not time but bandwidth. $\square$

5.2 Convergence Theorem

Theorem 15 (Dual-Pressure Convergence). *Let $\mathcal{S}$ be an adaptive system in an environment $\mathcal{E}$ satisfying either:*

- (a) $\tau > 0$, $H_\tau > 0$, and $\kappa > 0$ (temporal pressure), or
- (b) $\mathcal{D} > C$ and $\kappa > 0$ (informational pressure).

Then long-term persistence ($\lim_{n \rightarrow \infty} P(\text{survive } n \text{ steps}) > 0$) requires:

$$\mathcal{I} = I(X; Y_{t+\tau} | Y_t) > 0. \quad (11)$$

That is, the system must be predictive.

Proof. Under condition (a), the result follows from Theorem 8: a reactive system ($\mathcal{I} = 0$) faces exponential survival decay.

Under condition (b), the result follows from Theorem 13: a reactive system incurs information loss $\mathcal{L} > 0$, which produces action–state mismatches with compounding viability costs. To reduce information loss to survivable levels, the system must supplement channel-transmitted information with internally generated predictions, requiring $I(X; Y_{t+\tau} | Y_t) > 0$.

Since (a) and (b) are independent (Proposition 14) and each individually requires $\mathcal{I} > 0$, the conclusion holds under either condition alone and *a fortiori* when both obtain. $\square$

5.3 The Significance of Convergence

When two independent arguments, starting from different premises and employing different mathematical machinery, arrive at the same conclusion, this provides some evidence that the conclusion reflects a structural feature of the problem rather than an artifact of a particular theoretical lens.

A cautionary note is in order, however. The two pressures are not as deeply independent as they might first appear: both arise from the same underlying physical

reality—that information processing in material systems is finite and bounded. The temporal pressure stems from finite signal speed; the informational pressure stems from finite channel capacity. Both are consequences of the physical nature of computation. The convergence is therefore less surprising than it would be if the two arguments proceeded from genuinely unrelated domains (as, say, biogeographic and genetic evidence for evolution do). The convergence is better understood as *two facets of a single physical constraint*—the finitude of information processing—manifesting through different parameters (τ and C) and requiring the same architectural response.

Similarly, the temporal and informational arguments for predictive architecture are logically independent in the formal sense (Proposition 14)—one concerns time, the other concerns complexity—yet converge on the same architectural requirement. This convergence suggests that the need for predictive capacity is robust: it does not depend on a single parameter being large, but arises whenever *either* of two common physical constraints is binding.

Recent work in computational neuroscience provides further convergent evidence. Perinet [17] has demonstrated that heterogeneous synaptic delays in spiking neural networks can be *learned* to detect precise spatiotemporal spiking motifs. In that model, delays are not merely obstacles to be compensated but computational resources that encode temporal structure. The capacity to simultaneously learn weights and delays [17] shows that the temporal structure created by delays becomes substrate for the very predictions that the delays necessitate—a striking instance of constraint becoming resource.

6 A Toy Model: The Ring Environment

To make the formal results concrete and to demonstrate the phase boundary between reactive and predictive regimes, we construct a minimal dynamical model with fully specified parameters. The model is deliberately simple—a “spherical cow” designed not to capture the full richness of biological or artificial systems, but to exhibit the structural features our theorems identify: exponential survival decay for reactive agents, the dependence of this decay on volatility and channel capacity, and the rescue provided by predictive architecture.

6.1 Model Specification

Definition 16 (Ring Environment). The environment consists of L discrete states arranged on a ring (i.e., $Y = \mathbb{Z}/L\mathbb{Z}$, the integers modulo L). At each discrete time step,

the environment transitions according to:

$$Y_{t+1} = \begin{cases} Y_t & \text{with probability } 1 - q, \\ (Y_t + 1) \bmod L & \text{with probability } q/2, \\ (Y_t - 1) \bmod L & \text{with probability } q/2, \end{cases} \quad (12)$$

where $q \in [0, 1]$ is the *volatility parameter*.

This defines a symmetric random walk on a ring. The parameter q controls how rapidly the environment changes: at $q = 0$ the environment is static, at $q = 1$ it moves at every step. The ring topology ensures a finite, well-defined state space without boundary effects.

The conditional entropy of this process—the environmental volatility as defined in Definition 3—can be computed exactly for a single step:

$$H_1 = -q \log_2 \left(\frac{q}{2} \right) - (1 - q) \log_2 (1 - q). \quad (13)$$

For τ steps, the effective volatility H_τ grows with τ as the random walk disperses, approaching $\log_2 L$ (uniform uncertainty) as $\tau \rightarrow \infty$.

The agent observes the environment with a delay of τ steps and transmits its observation through a channel of capacity C bits per step. If $C \geq \log_2 L$, the full state can be transmitted without loss. If $C < \log_2 L$, the observation must be quantized: the L states are binned into 2^C levels, and the agent receives only the bin index—a lossy representation of the true state.

At each step, the agent must output an action (a predicted state). If the agent’s prediction differs from the actual environmental state by more than a tolerance of $\lfloor L/20 \rfloor$ on the ring, an error occurs. Each error is fatal with probability $\kappa = 0.3$.

Two agents are compared:

- **Reactive agent** ($\mathcal{I} = 0$): Outputs the (possibly quantized) delayed observation as its prediction. It has no model of the environment’s dynamics.
- **Predictive agent** ($\mathcal{I} > 0$): Estimates the environment’s velocity from consecutive delayed observations and extrapolates forward by τ steps. This corresponds to maintaining a first-order generative model: $\hat{Y}_{t+\tau} = Y_{t-\tau} + \hat{v} \cdot \tau$, where $\hat{v}$ is the estimated velocity.

This setup instantiates both pressures from the formal framework: the latency τ creates the temporal pressure, and the state-space size L relative to channel capacity C creates the informational pressure.

6.2 Analytical Predictions

The formal framework makes three specific predictions for this model, which we verify by simulation.

Prediction 1 (Exponential decay): For the reactive agent, survival probability should decay exponentially as $(1 - p_{\text{err}})^n$, where p_{err} depends on q , τ , κ , and C as specified in Proposition 7. This follows directly from Theorem 8.

Prediction 2 (Temporal pressure): At fixed L and full channel capacity ($C \geq \log_2 L$), increasing volatility q should accelerate the decay for the reactive agent while the predictive agent should show greater resilience. The difference in decay rates corresponds to the predictive capacity $\mathcal{I}$.

Prediction 3 (Informational pressure): At fixed q and τ , reducing channel capacity C below $\log_2 L$ should produce additional survival decay for the reactive agent—even at moderate volatility—because the quantized observation degrades the agent’s state estimate.

Prediction 4 (Phase boundary): In the joint (q, L) parameter space, the boundary between viable reactive regimes (long lifetimes) and nonviable regimes (rapid extinction) should shift: adding informational pressure (by limiting C) should shrink the viable region, while switching to a predictive agent should expand it.

6.3 Simulation Results

We simulated each agent configuration across parameter sweeps with $\tau = 3$ and $\kappa = 0.3$. The number of independent trials and time steps varied by analysis: 2000 trials of 200 steps for the survival curve analyses (Figures 1 and 2), 5000 trials of 300 steps for the exponential decay validation (Figure 3), 1000 trials of 300 steps for the advantage ratio analysis (Figure 5), and 500 trials of 300 steps per parameter combination for the phase diagram (Figure 4).

6.3.1 Temporal Pressure

Figure 1 shows survival curves for the reactive and predictive agents at fixed $L = 20$ with full channel capacity, across five volatility levels. The reactive agent (left panel) exhibits the predicted exponential decay, with the rate of decay increasing sharply with volatility: at $q = 0.9$, fewer than 5% of reactive agents survive 25 steps, while at $q = 0.1$, roughly 20% survive 200 steps. The predictive agent (right panel) shows substantially slower decay at the same volatility levels, confirming that the velocity-based model provides genuine delay compensation.

Temporal Pressure: Survival Under Increasing Volatility
($L = 20$, $\tau = 3$, $\kappa = 0.3$, full channel capacity)

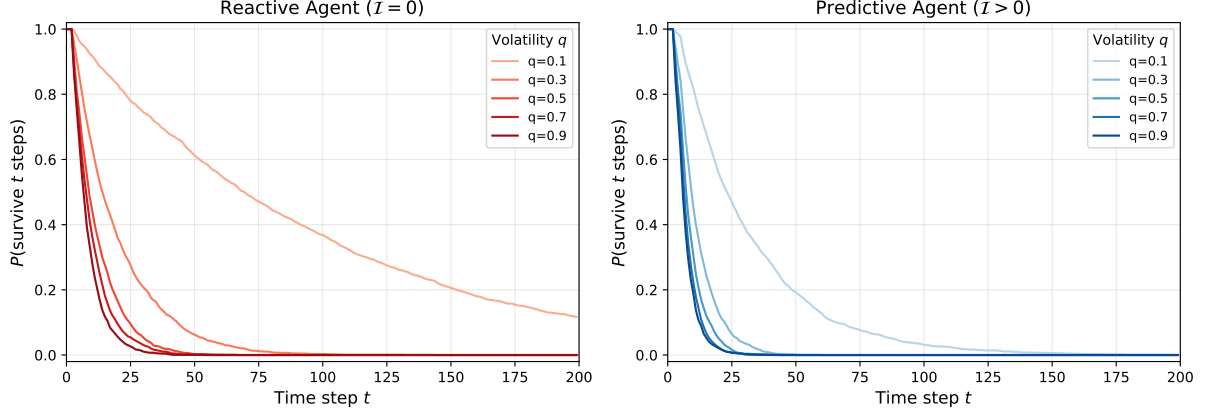


Figure 1: Survival curves under the temporal pressure. Left: reactive agent. Right: predictive agent. Full channel capacity ($C \geq \log_2 L$), so only the temporal pressure operates. Parameters: $L = 20$, $\tau = 3$, $\kappa = 0.3$.

6.3.2 Informational Pressure

Figure 2 isolates the informational pressure by fixing volatility at $q = 0.5$ and varying channel capacity from $C = 1$ bit to $C = 6$ bits ($= \log_2 64$, full capacity) with $L = 64$. For the reactive agent (left panel), reducing channel capacity produces a dramatic acceleration of extinction, even at moderate volatility. With only 1 bit of channel capacity, the reactive agent cannot distinguish most environmental states, and survival collapses rapidly. The predictive agent (right panel) also suffers at very low channel capacities but degrades more gracefully, because its model partially compensates for the lost information.

Informational Pressure: Survival Under Decreasing Channel Capacity
 $(L = 64, q = 0.5, \tau = 3, \kappa = 0.3, \log_2 L = 6 \text{ bits})$

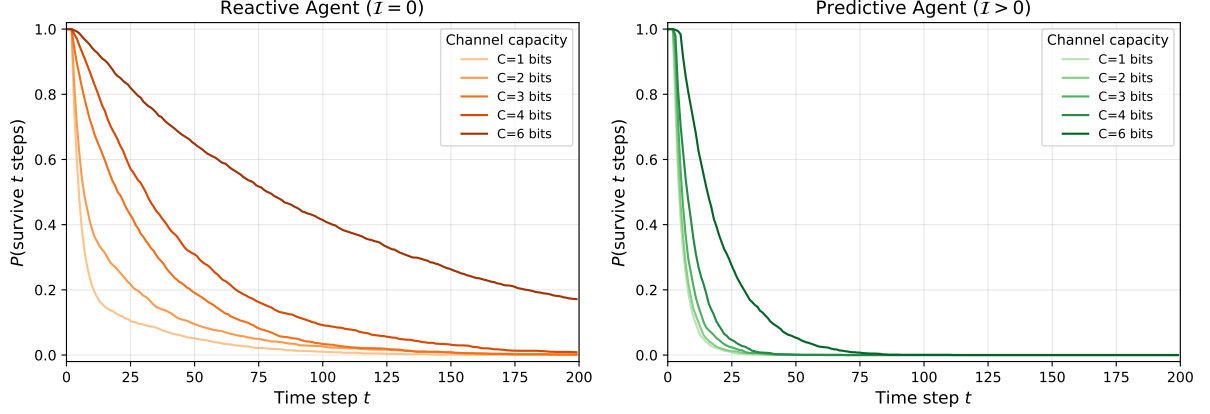


Figure 2: Survival curves under the informational pressure. Left: reactive agent. Right: predictive agent. Volatility fixed at $q = 0.5$. Decreasing channel capacity C below $\log_2 L = 6$ bits progressively degrades survival. Parameters: $L = 64, \tau = 3, \kappa = 0.3$.

6.3.3 Exponential Decay Validation

Figure 3 verifies that the simulated survival curves follow the exponential form $(1 - p_{\text{err}})^n$ predicted by Theorem 8. On a logarithmic vertical axis, the simulated curves (solid lines) align closely with the theoretical bound (dashed lines) across four volatility levels. The agreement confirms that the survival dynamics of the reactive agent are well described by the simple geometric decay model from our formal framework.

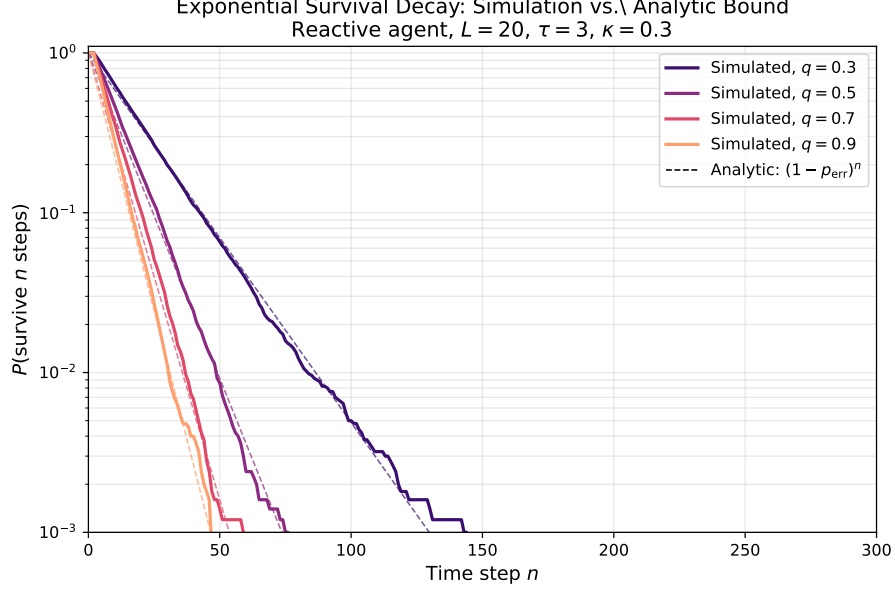


Figure 3: Exponential survival decay: simulation vs. analytic bound. Solid lines show simulated survival of the reactive agent on a log scale. Dashed lines show the fitted analytic bound $(1 - p_{\text{err}})^n$. The close agreement validates Theorem 8. Parameters: $L = 20$, $\tau = 3$, $\kappa = 0.3$, full channel capacity.

6.3.4 Phase Diagram

Figure 4 displays the full phase diagram in the (q, L) parameter space. Panel (a) shows the reactive agent with full channel capacity (temporal pressure only): the viable region (green, long lifetimes) occupies the low-volatility, low-complexity corner, and the phase boundary (dashed line) curves through the parameter space. Panel (b) shows the same reactive agent but with channel capacity limited to $C = 3$ bits: the viable region shrinks dramatically, because the informational pressure now compounds the temporal pressure. Panel (c) shows the predictive agent with the same limited channel: the viable region partially recovers, demonstrating that prediction compensates for both pressures simultaneously.

Phase Diagram: Mean Agent Lifetime ($\tau = 3, \kappa = 0.3$)
Dashed line: approximate phase boundary (mean lifetime = 150 steps)

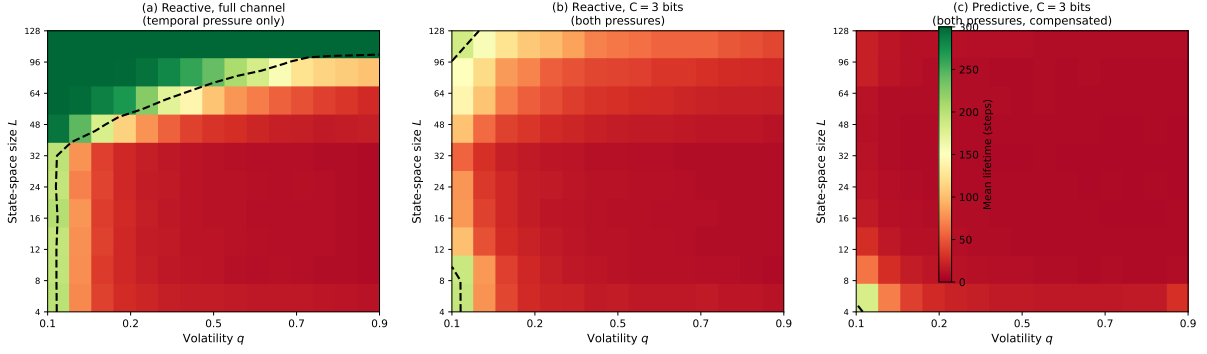


Figure 4: Phase diagrams showing mean agent lifetime across volatility q (horizontal axis) and state-space size L (vertical axis). Green indicates long survival; red indicates rapid extinction. Dashed lines mark the approximate phase boundary (mean lifetime = 150 steps). (a) Reactive agent, full channel: only temporal pressure operates. (b) Reactive agent, $C = 3$ bits: both pressures operate, viable region shrinks. (c) Predictive agent, $C = 3$ bits: prediction partially restores viability. Parameters: $\tau = 3, \kappa = 0.3$.

6.3.5 Predictive Advantage

Figure 5 quantifies the predictive advantage—the ratio of predictive to reactive mean lifetime—across volatility, for several channel capacities ($L = 48$). Two patterns emerge. First, the advantage increases with volatility: in slowly changing environments, prediction offers little benefit over reaction (both agents perform similarly), while in rapidly changing environments, the predictive agent substantially outlives the reactive one. Second, the advantage is greater when channel capacity is low: at $C = 2$ bits, the predictive agent achieves the largest relative advantage, because the informational pressure makes prediction’s compressive function most valuable precisely where raw transmission is most degraded.

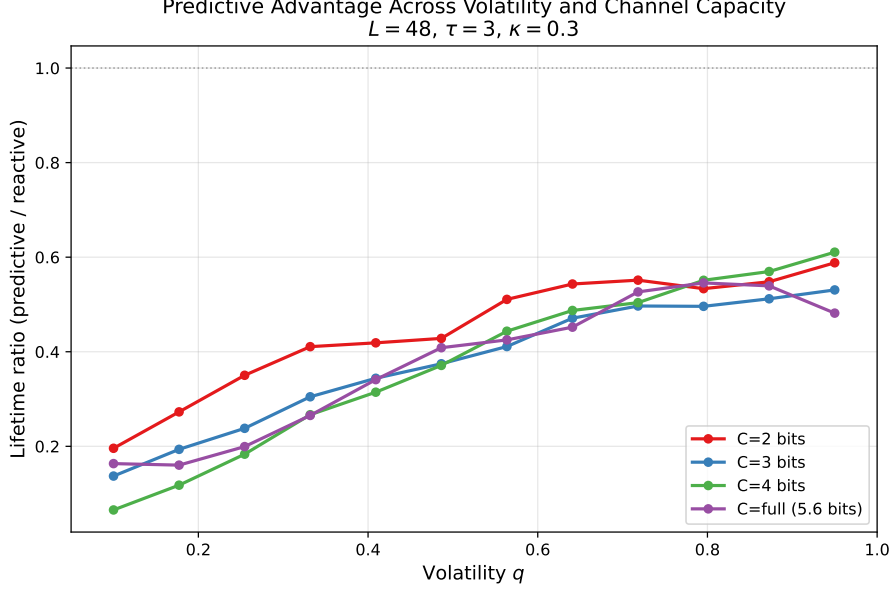


Figure 5: Predictive advantage (ratio of predictive to reactive mean lifetime) as a function of volatility q , for several channel capacities. The advantage increases with volatility and is greatest at low channel capacity, consistent with the dual-pressure framework. Parameters: $L = 48$, $\tau = 3$, $\kappa = 0.3$.

6.4 Summary of Toy Model Results

The simulation confirms all four analytical predictions:

1. Reactive survival decays exponentially, matching the $(1 - p_{\text{err}})^n$ bound from Theorem 8.
2. Increasing volatility accelerates reactive decay; prediction provides substantial rescue.
3. Reducing channel capacity below $\log_2 L$ produces additional decay via the informational pressure, independent of the temporal pressure.
4. The phase boundary between viable and nonviable regimes shifts predictably with both pressures, and prediction partially restores viability in regimes where reactive control fails.

The toy model thus provides a concrete demonstration that the formal theorems are not merely mathematical abstractions but describe observable dynamics in a specified system with known parameters.

A candid assessment of the model’s limitations is warranted. The ring environment with nearest-neighbor transitions is one of the simplest possible stochastic processes—a symmetric random walk. The predictive agent uses a first-order velocity estimate, which is well matched to this particular environment. The demonstrated predictive advantage is therefore partly an artifact of this match: in environments with more complex dynamics (nonlinear, multi-modal, or long-range-dependent), the required model complexity would

grow correspondingly, and a first-order velocity model would perform poorly. The toy model demonstrates the *existence* of the phase boundary and the *qualitative* behavior predicted by the theorems; it does not claim to represent the quantitative structure of the boundary in realistic ecological settings.

6.4.1 Sensitivity to the Punishment Parameter κ

The punishment parameter $\kappa = 0.3$ was held fixed in the preceding analyses. However, the formal framework predicts that the phase boundary location depends on κ : from Proposition 7, the per-step error rate scales as $p_{\text{err}} \propto \kappa \cdot H_\tau / \log |Y|$. For the ring environment, $H_1 \approx q$ for small q , so the critical volatility at which a reactive agent’s expected lifetime drops to some threshold n_{crit} scales approximately as:

$$q_{\text{crit}} \sim \frac{\log |Y|}{\kappa \cdot n_{\text{crit}}}, \quad (14)$$

indicating that $q_{\text{crit}} \propto 1/\kappa$: as punishment becomes more severe (larger κ), a lower volatility suffices to make reactive control nonviable. Conversely, as $\kappa \rightarrow 0$, the phase boundary recedes toward $q = 1$ —errors become inconsequential and the reactive regime expands to cover nearly all environments.

Figure 6 verifies this scaling. Panel (a) shows that the reactive agent’s mean lifetime is strongly modulated by κ : at $\kappa = 0.05$, the agent survives substantially longer across all volatility levels than at $\kappa = 1.0$. Panel (b) plots the critical volatility (defined as the q at which mean lifetime crosses 100 steps) against $1/\kappa$, showing a positive relationship consistent with the predicted scaling, though with sublinear growth at large $1/\kappa$ due to finite-size effects in the ring.

Sensitivity of Phase Boundary to Punishment Parameter κ
Reactive agent, $L = 20$, $\tau = 3$

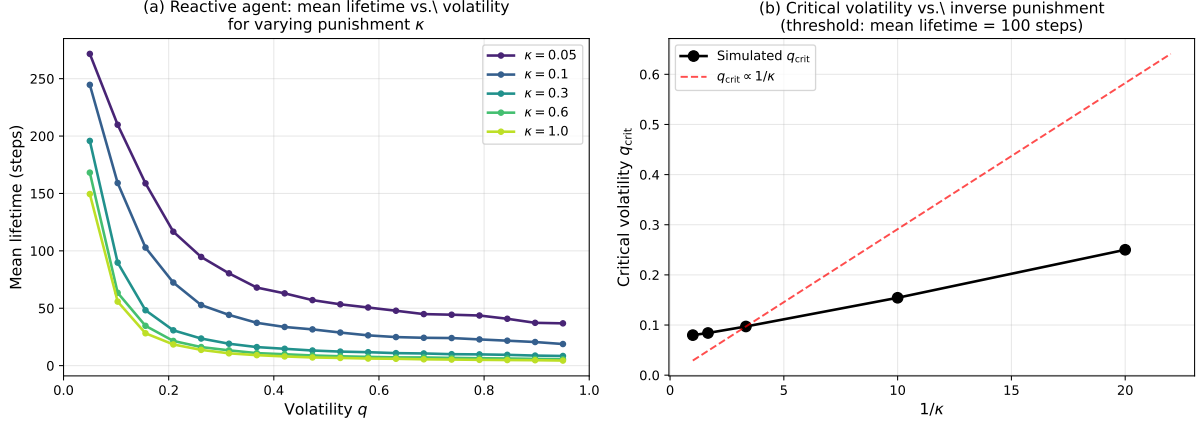


Figure 6: Sensitivity of the phase boundary to the punishment parameter κ . (a) Mean lifetime of the reactive agent as a function of volatility q , for five values of κ . (b) Critical volatility q_{crit} (threshold: mean lifetime = 100 steps) as a function of $1/\kappa$. The dashed line shows the theoretical scaling $q_{\text{crit}} \propto 1/\kappa$ from (14). Parameters: $L = 20$, $\tau = 3$, full channel capacity.

7 The Transition from Reactive to Predictive: A Phase Boundary

The arguments above establish that prediction is necessary under certain conditions. But where exactly does the transition occur? We can characterize this transition as a phase boundary in the space of system–environment parameters.

7.1 The Reactive Regime

A purely reactive architecture is viable when both pressures are weak:

$$\kappa \cdot \frac{H_\tau}{\log |Y|} \ll 1 \quad \text{and} \quad \mathcal{D} \leq C. \quad (15)$$

These conditions describe small organisms in slow, simple, forgiving environments. The nematode *C. elegans*, with 302 neurons spanning a body of approximately 1 mm, operates in this regime: conduction delays are negligible, the environment changes slowly, and errors are rarely fatal [24].

7.2 The Predictive Regime

The predictive regime obtains when either pressure becomes strong:

$$\kappa \cdot \frac{H_\tau}{\log |Y|} \gtrsim \epsilon_{\text{crit}} \quad \text{or} \quad \mathcal{D} \gg C, \quad (16)$$

where ϵ_{crit} is the maximum tolerable per-step error rate. In this regime, reactive systems face exponential survival decay, and only predictive architectures persist.

7.3 The Dual-Pressure Region

Most ecologically relevant scenarios—large organisms hunting mobile prey in complex environments—fall in a region where *both* pressures are active simultaneously. Here, the system faces both the latency gap and the bandwidth bottleneck, and prediction serves dual functions: temporal compensation and informational compression.

This dual functionality helps explain a feature of biological neural architecture: the same structures implicated in temporal prediction (cerebellar forward models, hippocampal predictive coding) also perform data compression and pattern extraction. From the dual-pressure perspective, this is expected—a single predictive model simultaneously solves both problems.

8 Connections to Neuroscience

8.1 Neural Delays as the Temporal Pressure in Action

The biological nervous system provides a concrete instantiation of the temporal pressure. Neural signals in the fastest myelinated axons propagate at approximately 100 m/s [23]. For a signal to travel from the retina through the lateral geniculate nucleus to primary visual cortex, 50–150 ms elapse. Adding motor planning and execution, total sensorimotor loop latencies reach hundreds of milliseconds.

Perrinet et al. [18] demonstrated this quantitatively through neuronal simulations of smooth pursuit eye movements. In their active inference model, uncompensated delays produce exponentially growing oscillations that render oculomotor tracking impossible. The solution—representing target dynamics in generalised coordinates of motion—enables the system to project delayed sensory signals forward in time, yielding stable pursuit.

The perceptual manifestation of delay compensation is visible in the flash-lag effect [8]. Khoei et al. [11] formalized this within a Bayesian framework, showing that a motion-based predictive model—termed *parodiction*—accounts for both the standard flash-lag effect and the conditions under which it disappears.

8.2 Synaptic Delays as Learnable Parameters: From Constraint to Mechanism

The informational pressure manifests at the neuronal level in a way that transforms an apparent obstacle into a computational resource. A standard treatment of synaptic delays regards them as fixed biophysical properties—axonal length, myelination thickness, synaptic vesicle release kinetics—that the nervous system must tolerate. The dual-pressure framework suggests a different interpretation: delays are not merely tolerated but *tuned*, and their heterogeneity across a neural population may constitute a structural substrate for predictive compression.

8.2.1 The HD-SNN Model: Delays as Learnable Degrees of Freedom

Perrinet [17] introduced a Heterogeneous Delays Spiking Neural Network (HD-SNN) in which both synaptic weights and delays are learnable parameters. Each synapse s in the network is defined by a tuple $(a_s, b_s, w_s, \delta_s)$: a presynaptic address a_s , a postsynaptic address b_s , a weight w_s , and a delay $\delta_s \in \{0, 1, \dots, D - 1\}$. The key innovation is that the delay δ_s is not frozen but participates in gradient-based learning alongside the weight w_s .

The detection of a spiking motif at postsynaptic neuron b and time t is computed as:

$$p(b, t) = \sigma \left(K_B(b) + \sum_{a \in \mathcal{S}^b} \sum_{0 \leq \delta \leq D} A(a, t - \delta) \cdot K^b(a, \delta) \right), \quad (17)$$

where $A \in \{0, 1\}^{N \times T}$ is the observed raster plot, $K^b \in \mathbb{R}^{N \times D}$ is the receptive field kernel for neuron b , and σ is the sigmoid function. This takes the form of a temporal convolution with learnable delay structure. The model thus inverts a generative process: a high-dimensional raster plot (many neurons, many time steps) is decomposed into a small number of identified motifs—each carrying compact, predictive information about the stimulus.

A result of particular relevance is that, in Perrinet’s model, the simultaneous learning of weights *and* delays substantially outperforms architectures where delays are fixed (as in the classical polychronization model of Izhikevich 9). Detection accuracy increases significantly with the temporal depth D of the delay kernel, demonstrating quantitatively that heterogeneous delays provide a computational advantage that grows with the richness of the temporal structure being encoded.

This result has a natural interpretation within the dual-pressure framework, though one that requires careful statement. A spiking neural network that can only adjust weights has access to a parameter space of dimension $N \times M$ (weights connecting N presynaptic to M postsynaptic neurons). A network that can also adjust delays expands this to

$N \times M \times D$ —a much richer space of possible input–output mappings. The additional degrees of freedom provided by learnable delays allow the network to represent a larger set of temporal patterns.

An important distinction should be drawn here. In the formal framework, C denotes the raw channel capacity of the sensor-to-action pathway—a fixed physical quantity determined by the number of neurons, their firing rates, and the noise characteristics of synaptic transmission. Learnable delays do not increase this raw capacity; the same spikes traverse the same axons. What learnable delays increase is the *efficiency* with which the available capacity is used: by allowing the decoder to align its temporal filters with the temporal structure of the signal, learnable delays reduce the effective information loss in the decoding process. In information-theoretic terms, learnable delays move the system closer to the channel capacity bound without increasing the bound itself. The effect is analogous to optimizing a codebook in communication theory: the channel capacity remains fixed, but better codes extract more of the available capacity. For the informational pressure argument, this distinction matters: learnable delays help the system cope with the pressure by making better use of a fixed channel, not by relaxing the channel constraint.

8.2.2 Delay Distributions as Encodings of Temporal Derivatives

There is a suggestive connection between heterogeneous delay distributions and the *generalised coordinates of motion* employed in active inference models [5, 18]. In the active inference framework, the system represents not only the current value of a sensory signal but also its first derivative (velocity), second derivative (acceleration), and higher temporal derivatives. These representations in generalised coordinates enable the system to project delayed sensory signals forward in time, compensating for latency.

A population of synapses with heterogeneous delays $\delta_1 < \delta_2 < \dots < \delta_D$ connecting the same presynaptic neuron to a postsynaptic target effectively provides the postsynaptic neuron with a temporal sampling of the presynaptic signal at D different time offsets. From this temporal sample, finite-difference approximations to the signal’s derivatives can, in principle, be extracted. A pair of synapses with delays δ_1 and δ_2 provides information proportional to the first temporal derivative (velocity) of the presynaptic signal; a triplet provides an approximation to the second derivative (acceleration); and so on.

This suggests a possible structural interpretation: the distribution of synaptic delays in a neural population may function as a biological implementation of generalised coordinate encoding. Where the active inference framework posits abstract representations of temporal derivatives, the HD-SNN model offers a concrete biophysical mechanism—heterogeneous delays—through which such representations could be physically realized. Each delay value acts as a temporal tap, and the learned weight at that delay determines

how strongly the signal at that time offset contributes to the postsynaptic response. The full delay kernel $K^b(a, \delta)$ thus encodes a learned temporal filter that may approximate the derivative structure needed for prediction.

We emphasize that this interpretation remains speculative; the formal equivalence between delay-based temporal sampling and generalised coordinate representation has not, to our knowledge, been rigorously established. It is offered here as a hypothesis that may merit investigation.

8.2.3 Delay Heterogeneity and Channel Capacity

The connection between learnable delays and the informational pressure can be made more precise. Consider a neural pathway that must transmit information about a time-varying stimulus through a population of N neurons, each firing at a maximum rate of r spikes per second. The raw channel capacity of this pathway is bounded by $C \leq N \cdot r \cdot \log_2(1/r\Delta t)$ bits per second, where Δt is the temporal resolution.

If all synapses have identical delays, the postsynaptic neuron receives a single temporal snapshot of the presynaptic population—a spatial pattern at one time point. The information carried is limited to the spatial structure of the firing pattern at that instant.

If synapses have heterogeneous delays spanning a range $[0, D\Delta t]$, the postsynaptic neuron receives a spatiotemporal volume of information: the firing pattern across D time bins. The effective information available to the postsynaptic neuron can in principle increase with D , the temporal depth—but the magnitude of this gain depends on the temporal correlation structure of the presynaptic signal. For signals with rich temporal structure (strong autocorrelations, characteristic motifs), successive time bins carry substantially different information, and the gain from temporal depth may approach a factor of D . For signals with flat temporal spectra (temporally uncorrelated firing), successive time bins are nearly independent samples of the same distribution, and additional delay taps provide diminishing returns. The gain from delay heterogeneity is therefore signal-dependent, and the factor D represents an upper bound that is approached only when the signal’s temporal structure is rich relative to the delay range. In this qualified sense, delay heterogeneity functions as a *temporal buffer* that redistributes information across time, allowing the postsynaptic neuron to integrate evidence from a wider temporal window without requiring additional neurons or higher firing rates.

From the perspective of the informational pressure, this is significant. When the reactive transmission demand $\mathcal{D}$ exceeds the instantaneous channel capacity, delay heterogeneity provides a mechanism for temporal multiplexing: information that cannot be transmitted in a single time step is spread across multiple steps and reassembled at the postsynaptic site through the learned delay structure. This suggests that delay heterogeneity may function as a structural substrate for predictive compression—a biophysical

mechanism through which the nervous system implements the model-based coding strategy that our formal framework identifies as necessary under informational pressure.

8.3 Two Asymmetric Cases: Latency Without Complexity, Complexity Without Latency

To guard against the impression that the dual-pressure framework reduces to a theory of delay compensation, it is worth examining two limiting cases in which the pressures are decoupled.

High complexity, negligible latency. Consider a hypothetical system with near-instantaneous response ($\tau \approx 0$) operating in an environment of high algorithmic complexity. A small insect’s nervous system, spanning fractions of a millimeter, approximates this regime: conduction delays are negligible, but the visual environment (a forest floor, a turbulent stream) may present stimulus patterns of considerable Kolmogorov complexity. In such a system, the temporal pressure is weak, yet the informational pressure remains strong. The system cannot transmit the full complexity of the visual scene through its limited neural channel. It must compress, and the only compression that preserves action-relevant information is predictive: encoding the regularities (edges, textures, motion patterns) and transmitting only the residuals. This case suggests that even organisms with minimal neural latency may require predictive architectures—not to compensate for delay, but to manage complexity.

High latency, low complexity. Conversely, consider a large organism in a simple, slowly changing environment. A large tortoise in a stable grassland faces significant conduction delays (body length of a meter, round-trip latencies of tens of milliseconds) but low environmental complexity: the grass does not move unpredictably, threats are rare and slow. Here the temporal pressure exists in principle but is weak in practice, because the environment changes little within the delay window ($H_\tau \approx 0$). The informational pressure is also minimal ($\mathcal{D} \ll C$). In this regime, the framework correctly predicts that predictive architecture, while present, may be minimal—limited to basic proprioceptive forward models for locomotion rather than elaborate anticipatory world models.

These asymmetric cases demonstrate that the dual-pressure framework does not reduce to a single-variable theory. Latency alone does not mandate elaborate prediction if the environment is simple. Complexity alone mandates prediction even if latency is negligible. The richest predictive architectures—those seen in large-brained predators hunting mobile prey in complex environments—arise where both pressures converge.

8.4 The Free Energy Principle

The free energy principle [5] provides a comprehensive framework within which both pressures can be situated. Under this principle, adaptive systems minimize variational free energy—an information-theoretic quantity that bounds surprise. Minimizing free energy entails maintaining a generative model of the environment’s causal structure, which is the predictive architecture our arguments identify as necessary.

The present paper complements the free energy principle by providing two specific *physical reasons why* free energy minimization is not merely advantageous but mandatory. The temporal pressure shows that without prediction, latency produces fatal errors. The informational pressure shows that without prediction, complexity overwhelms the channel. The free energy principle describes what the system does; our arguments explain why it has no choice.

9 Implications for Artificial Intelligence

9.1 Prediction in Machine Learning

Modern AI systems have independently converged on predictive architectures. Large language models are next-token prediction engines [20]. Reinforcement learning agents learn world models that predict state transitions [7]. The dual-pressure framework suggests this convergence reflects a structural necessity, not merely a convenient training objective.

9.2 Prediction Errors and the Question of Hallucination

A predictive system that compresses environmental information into an internal model will sometimes generate outputs that diverge from the true environmental state. In AI, such divergences are often called “hallucinations”; in biological perception, analogous phenomena include the flash-lag effect, visual filling-in, and phantom limb sensations.

It is tempting to claim that hallucination is a “necessary cost” of prediction under compression. The dual-pressure framework supports a more circumscribed version of this claim: compression under uncertainty *guarantees that prediction errors will occur with nonzero probability*. A model that encodes only the predictable regularities ρ_{pred} and discards the stochastic component ρ_{stoch} will, by construction, fail to track genuinely random environmental fluctuations. When the model fills in the discarded component with its best estimate, that estimate will sometimes be wrong.

However, several qualifications are needed. First, not all prediction errors are “hallucinations” in the same sense. A visual system that extrapolates a moving object half a degree ahead of its true position (the flash-lag effect) commits a small, systematic, computationally motivated error. A language model that fluently asserts a false historical

fact commits a qualitatively different kind of error—one involving confident confabulation about structured knowledge. Compression under uncertainty produces the first kind of error; the second may involve additional failure modes (distributional drift, out-of-distribution extrapolation, training data biases) that are not fully captured by the informational pressure alone.

Second, the claim that eliminating hallucination requires “zero latency or infinite channel capacity” is too strong as stated. There exist error-reduction strategies—retrieval augmentation, output verification, uncertainty quantification, constrained decoding—that can substantially reduce certain classes of prediction error without modifying the system’s fundamental latency or channel capacity. What the framework does support is the weaker claim that *some* irreducible prediction error remains for any finite-capacity system operating in a partially unpredictable environment, because the stochastic component $\rho_{\text{stoch}} > 0$ is by definition beyond the reach of any model. The magnitude and character of this irreducible error depend on the specific environment and model class, and characterizing it precisely remains an open problem.

9.3 Predictions for Future AI Development

The framework generates several specific predictions:

1. AI systems deployed in real-time physical environments with significant latency will converge on predictive architectures regardless of their initial design.
2. The complexity of the internal model an AI system develops will scale with both the volatility and the Kolmogorov complexity of its operating environment.
3. Some irreducible prediction error will persist in any AI system operating under finite capacity in partially unpredictable environments. However, specific classes of error (confident confabulation, factual hallucination) may be substantially reducible through architectural interventions that do not require eliminating compression itself.

10 Objections and Boundary Cases

10.1 Is the Informational Argument Merely Shannon’s Theorem Restated?

The contribution is not the theorem but its application and its convergence with the temporal argument. Shannon’s theorem tells us that a channel of capacity C cannot transmit at rate $\mathcal{D} > C$. Our argument uses this to show that *adaptive* systems under bandwidth constraints must build internal predictive models. This depends on the additional premise that the system must act correctly in a punishing environment, which

transforms an information-theoretic constraint into an architectural mandate.

10.2 Is Kolmogorov Complexity Computable?

Kolmogorov complexity is uncomputable in the formal sense [15]. This does not undermine the argument, because the argument does not require computing $K(s)$. It requires only the *existence* of a gap between the complexity rate ρ and the channel capacity C . The uncomputability of K means that the system cannot determine the optimal compression, but this is precisely why evolution and learning are necessary—they approximate, through trial and error, a compression that no algorithm could compute exactly.

However, a more substantive concern deserves attention. The entire informational argument is built on ρ , which involves the asymptotic Kolmogorov complexity rate—a quantity that is not only uncomputable but also not robustly approximable for finite sequences in general. The argument’s empirical implications therefore depend on replacing K with computable proxies. Several such proxies exist and are routinely used in practice: statistical complexity [3], Lempel-Ziv compression ratios, prediction-error rates of fitted models, and the entropy rate of learned probabilistic models. Each of these approximates aspects of Kolmogorov complexity from above, and for the purposes of the informational argument, an upper bound on ρ suffices: if even the computable approximation exceeds channel capacity, the informational pressure is genuine. The framework should be understood as using Kolmogorov complexity as a theoretical foundation, with the understanding that empirical applications will substitute computable approximations.

10.3 What About Fully Random Environments?

If $\rho_{\text{pred}} = 0$, no internal model can help, and the informational argument does not apply. However, fully random environments are also environments in which *no* adaptive strategy is possible. The informational argument applies in the interesting middle ground where environments are partially predictable: $0 < \rho_{\text{pred}} < \rho$.

10.4 Does the Argument Apply to Very Simple Organisms?

For very small organisms in simple environments, both pressures may be weak enough that reactive architectures suffice. The framework does not claim that all life is predictive; it claims that wherever the pressures are strong, prediction is mandatory.

10.5 Is the Definition of Predictive Capacity Too Broad?

A legitimate concern is that Definition 6—which requires only that $I(X; Y_{t+\tau} | Y_t) > 0$ —is broad enough to classify almost any memory-bearing system as “predictive.” A system

that simply stores past observations in a buffer, without explicitly modeling future states, may carry some mutual information with the future simply because past and future are correlated through the environment’s transition structure. Such a system would satisfy $\mathcal{I} > 0$ without implementing anything resembling a generative model.

This is a genuine feature of the framework, not a bug, but it demands transparent acknowledgment. The formal results establish that *purely memoryless* reactive systems ($\mathcal{I} = 0$) face exponential survival decay. They do not establish that a full hierarchical generative model is the only architecture that achieves $\mathcal{I} > 0$. Between the memoryless extreme and a complete world model lies a spectrum of architectures that carry varying amounts of predictive information:

- *Buffered systems* that store recent observations carry $\mathcal{I} > 0$ to the extent that past observations predict the future.
- *Cached-policy systems* that store learned stimulus–response mappings (habits) carry predictive information implicitly, embedded in the policy rather than in an explicit state estimate.
- *Reservoir and echo state networks* that maintain high-dimensional dynamical state may carry substantial $\mathcal{I}$ without explicit temporal modeling.
- *Error-correcting systems* that use redundancy to tolerate transmission errors achieve a form of robustness distinct from prediction.

The framework’s contribution is to establish that $\mathcal{I} = 0$ is nonviable under the stated conditions. Where a given system falls on the spectrum from minimal buffer to full generative model is determined by the specific magnitudes of the temporal and informational pressures. The stronger both pressures, the richer the predictive representation must be—but characterizing this quantitative relationship precisely is beyond the current formal treatment and constitutes an important direction for future work.

A related question is whether, from the framework’s perspective, different architectures achieving the same value of $\mathcal{I}$ are functionally equivalent. The formal results are stated purely in terms of the *magnitude* of the conditional mutual information $I(X; Y_{t+\tau} \mid Y_t)$; they do not distinguish between architectures that achieve a given $\mathcal{I}$ through different mechanisms. A system that maintains an explicit generative model of the environment’s dynamics and a system that achieves the same $\mathcal{I}$ through a learned habitual policy both satisfy the formal requirement equally.

However, there are reasons to suspect that the *form* of the predictive information matters beyond what the current framework captures. First, explicit state prediction supports flexible, context-dependent action selection (planning), while policy-embedded prediction supports only the specific actions for which the policy was trained. Second, explicit models are compositional—they can be recombined to address novel situations—while cached policies are not. Third, the informational pressure argument suggests that model-based compression (transmitting prediction errors rather than raw states) achieves greater band-

width savings than policy-based compression (which must still transmit enough state to select among cached policies). These considerations suggest that the framework, as currently formulated, captures a necessary condition ($\mathcal{I} > 0$) but does not fully characterize the *sufficient* architecture. A more refined framework would need to distinguish not only the magnitude of predictive information but also its representational format—a distinction that connects to the broader debate in cognitive science between model-based and model-free control. We note this as an important limitation of the present treatment and a direction for future theoretical development.

11 Discussion

11.1 The Delay Paradox: Constraint as Resource

A recurring theme in the neuroscience of prediction is what might be called the *delay paradox*: neural delays create the problem that prediction solves, but delays also create the temporal structure that prediction exploits. As discussed in Section 8.2, the heterogeneous delays in Perrinet’s HD-SNN model [17] do not merely impose a cost; they provide the temporal scaffolding on which spiking motif detection is built. The learned delay kernels function as temporal filters that decompose high-dimensional raster plots into compact motif representations—a form of compression that the informational pressure makes necessary. Similarly, the generalised coordinates in active inference models [18] extract information from the temporal derivatives that delays make observable.

The possibility raised in Section 8.2—that delay heterogeneity may serve as a structural substrate for predictive compression—suggests that the delay paradox is not paradoxical at all, but rather an instance of a broader principle: that physical constraints on information processing do not merely mandate prediction but also provide the material from which predictive mechanisms are built. If this interpretation is correct, then the search for the neural implementation of predictive processing should attend closely to the distribution of synaptic delays, not only to synaptic weights. The recent demonstration that delays can be learned alongside weights through gradient-based optimization [10, 17] opens a concrete experimental and computational path toward testing this hypothesis.

11.2 Relation to Existing Frameworks

The dual-pressure argument complements rather than replaces existing theoretical frameworks. *Predictive coding* [21] describes how cortical circuits implement prediction; our framework explains why such circuits are necessary. *Active inference* [5] describes how systems minimize free energy; our framework identifies the two physical reasons why this is mandated. *Forward models in motor control* [25] describe how the brain predicts

sensory consequences of actions; our framework shows these are instances of a general architectural requirement. *Ecological psychology* [6] emphasizes richly structured environments; our framework adds that directly available information still requires time to propagate and bandwidth to transmit.

11.3 Toward Information Physics

At the most general level, the dual-pressure argument suggests that predictive architecture is a consequence of the *information physics* of adaptive systems: the conjunction of finite signal speed, finite channel capacity, environmental change, and selection for viability. These are physical constraints, not biological ones. Any system that satisfies the premises of Theorem 15 is subject to the conclusion.

12 Conclusion

This paper has identified two logically independent pressures that each force adaptive systems toward predictive architectures:

1. **The temporal pressure:** Finite signal propagation speed creates a latency gap between the world’s current state and the system’s ability to respond. In volatile, punishing environments, this gap produces exponentially decaying survival probability for reactive systems, which can be mitigated only by maintaining internal states that carry information about the future.
2. **The informational pressure:** High algorithmic complexity of the environment creates a bandwidth bottleneck when the reactive transmission demand exceeds channel capacity. The only compression strategy that preserves action-relevant information is one that exploits predictive regularities—an internal model that transmits prediction errors rather than raw state descriptions.

These two pressures are independent: the first concerns time, the second concerns complexity. Neither entails the other. Yet both converge on the same conclusion: that purely reactive (memoryless) adaptive systems cannot persist, and that some form of predictive internal representation—internal states carrying information about the future—is required. The strength of the claim should be stated precisely: the framework rules out $\mathcal{I} = 0$ under the stated conditions, establishing a minimal requirement for predictive capacity. How richly that requirement is met—whether through simple memory buffers, cached policies, or full generative world models—depends on the specific magnitudes of the pressures and remains an important open question.

The framework connects to recent work on delay compensation in oculomotor control [18], Bayesian motion prediction in perception [11], learnable delays in spiking neural networks [17], the free energy principle [5], forward models in motor control [25], the

evolutionary elaboration of predictive architectures [19], and the physical constraint argument for prediction [13].

The world of direct perception, of exhaustive reasoning, of stimulus–response chains uncomplicated by anticipation—this world never existed and cannot exist. The physics and mathematics forbid it. Every brain that ever evolved and every artificial system that ever scaled was already, and always, predicting.

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